Mask-Augmented Multimodal Animal Behaviour Recognition

Does the animal's foreground silhouette *shape* help where background segmentation did not?

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The problem

- Recognising what an animal is *doing* from camera-trap clips matters for ecology and welfare.
- Modern benchmarks are **multimodal**: video, audio, and reference scene segmentation maps.
- On the **MammAlps** benchmark:
    - audio gives a small lift over video;
    - **background / scene segmentation did *not* help** behaviour recognition.

The gap

Telling the model about the static *background* does not help it read behaviour. What about the representation that throws the background away entirely?

The idea (and the novelty)

Use the animal's foreground *silhouette shape* as a fused, positive modality.

- Extract it with **SAM 2**, tracked across the clip; fuse it on top of video + audio.
- A silhouette's changing shape *is* posture; two animals' overlapping masks *is* contact.

Why it is new — nobody treats shape this way:

- MammAlps scene maps: *background*, shown not to help.
- PanAf-FGBG: masks used to *erase* the animal and study the background.
- MAROON: masks as background-removed cutouts, not shape as its own stream.

Hypothesis: shape helps **posture- and contact-defined** behaviours (e.g. chasing, courtship) where the background could not.

Closest work: same mask, *opposite* purpose

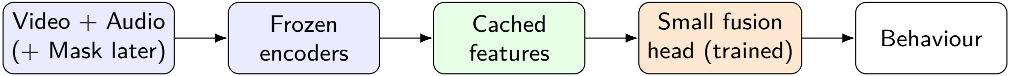
PanAf-FGBG is the nearest work — it also uses SAM 2 masks, but for the opposite reason:

	PanAf-FGBG	**This project**
Mask used to	remove / swap the background	derive the silhouette *shape*
Object of study	the *background* (bias)	the foreground *shape*
Fed to the model	RGB *pixels* (appearance)	*geometry* (outline, overlap)
Question	over-reliance on background?	does shape *add* signal?

- They cut the animal out to study *the room*; we keep only its changing *outline* — discarding the room *and* the appearance.
- Their finding (models over-rely on background) is *why* a background-free shape stream should be more robust.

Approach — deliberately lightweight

Heavy encoders are **frozen** and computed **once**; only a small fusion head is trained.



- Cheap to run; the same cached features serve every experiment.
- Capacity stays identical across modalities, so any gain reflects *information*, not parameters.
- The mask stream is **droppable at test time** (modality dropout).

The frozen building blocks (we *don't* train these)

Off-the-shelf **pretrained** models, run once and cached — never updated:

Modality	Pretrained model	Input → output
Video	VideoMAE (Kinetics)	16 frames → 768-d vector
Audio	AST (AudioSet)	wav → spectrogram → 768-d vector
Shape	SAM 2 + shape descriptors	frame + animal box → *mask* → shape numbers

- **SAM 2 is the odd one out:** a *segmentation* model — it outputs a *mask*, not an embedding. We turn the mask into shape numbers and a *tiny* temporal encoder summarises them over time.
- **Only the fusion head is trained** — the three big models never learn anything in this project.

Progress so far

- **Dataset secured & verified** — MammAlps Benchmark I; splits confirmed (train 4205 / val 686 / test 1244 / total 6135), matching the paper.
- **Scoring contract locked** — a dummy-prediction file passes the official evaluator's exact format check (the 1244-clip assertion).
- **Frozen encoder path working** — a real clip runs through VideoMAE and the audio encoder to give 768-d embeddings on the laptop GPU.
- **Reproducible codebase** — config-driven, tested, on GitHub (`mambr`); proposal and write-up scaffolding in place.

Foundations are in place; no behaviour model trained yet — that is next.

Plan / next steps

1. Cache features for all 6,135 clips on a Kaggle GPU (run once).
2. Reproduce the **video** and **video+audio** baseline (target $\approx$ 0.45 / 0.47 class-average mAP).
3. Add the **SAM 2 shape** modality and run the ablation:
   video $\rightarrow$ +mask $\rightarrow$ +audio $\rightarrow$ +audio+mask.
4. **Per-behaviour analysis** (chasing, courtship) and a background-shift robustness test.

Bar for success

A clean baseline, then: does shape help *where background did not*? Even a clean negative result is publishable.

Discussion

- Does the scope (single-animal MammAlps clips + a controlled multi-animal set for contact behaviours) look right?
- Any preference on the exact frozen encoders / compute?
- Happy to share the short proposal and the repository.

Thank you — questions & feedback welcome!